

4 (long)

- (a) Find the general solution to the differential equation

$$L\frac{d^2I}{dt^2} + R\frac{dI}{dt} + \frac{I}{C} = 0$$

describing the current response I of a resistor R , an inductor L , and a capacitor C connected in series for the following three cases:

- (i) $R^2 - 4\frac{L}{C} > 0$;
- (ii) $R^2 - 4\frac{L}{C} = 0$;
- (iii) $R^2 - 4\frac{L}{C} < 0$. [10]

- (b) Determine which case in (a) above gives an unstable solution for I . [5]

- (c) Show that the current response I is oscillatory for $R = 100 \, \Omega$, $L = 1 \times 10^{-2} \, \text{H}$, and $C = 2 \times 10^{-6} \, \text{F}$, and determine the characteristic frequency. [8]

- (d) If the initial conditions are $I = 1 \, \text{A}$ and $dI/dt = 0$ at $t = 0$, derive the complete solution for the current as a function of t . [7]